Chapter 1

Using AI for Ideation and Planning

IN THIS CHAPTER

» Coming up with ideas, with a little help from AI

» Dealing with AI hallucinations

» Walking through the process of AI ideation

» Picking the right AI tools to help your team get creative

Using artificial intelligence (AI) to help with creativity and ideation (coming up with new ideas), in particular, is not a new thought. In 2018, the AI-generated artwork "The Portrait of Edmond de Belamy" sold for $432,500 at Christie's auction house. This price was almost 45 times its pre-auction high estimate. This piece was created by an AI model trained on a data set of 15,000 portraits spanning six centuries.

The type of generative AI system used to create "The Portrait of Edmond de Belamy" was a *generative adversarial network* (GAN). GANs use two neural networks: one network that generates images and the other that evaluates them. GANs can produce hyper-realistic images (of people and places, for example) that are often indistinguishable from real ones. If

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enced GANs at work.

When you put the thought of creativity into the context of your business, ideation serves as a foundational exercise for the development of success-

ful marketing strategies, content, and campaigns. Ideation isn't just about tossing around ideas; it also involves the art of exploring your customer base, the quest to uncover possibilities for enhancing market success, and the identification of potential paths forward. And ideation doesn't happen in a vacuum. Customer research, business objectives, competitor dynamics, product plans, and previous marketing results all serve as the ingredients for the ideation process.

On the flip side, your marketing planning takes the diverse ideas developed through the ideation process and meticulously orchestrates them into actionable steps and strategies that sketch out a road map for achieving desired outcomes. These outcomes can take the form of an annual marketing plan, a campaign brief, a creative concept, or even a media plan.

The emergence of generative AI ushers in a new era of possibilities and efficiencies in how businesses can ideate, conceptualize, strategize, and develop marketing (and even new product) innovations. In this chapter, I discuss how you can use AI and, specifically, generative AI to fuel the ideation processes.

Engaging AI to Ideate on Behalf of Human Beings

Until recently, the notion that AI can match human ideation seemed fantastical. Conventional wisdom dictates that AI tools aren't sophisticated enough to generate ideas, develop names for products, identify solutions for unmet needs, or unpack creative insights.

To test this hypothesis, two University of Pennsylvania Wharton School of Business professors ran a controlled experiment to understand how an AI performed on an unstructured assignment relative to the professors' MBA students. This study was highlighted in a *Wall Street Journal* article, "M.B.A. Students vs. AI: Who Comes Up With More Innovative Ideas?" from September 9, 2023. See the article at `www.wsj.com/tech/ai/mba-students-vs-chatgpt-innovation-679edf3b` and the sidebar "ChatGPT versus Wharton MBA students" in this chapter for details of this study.

This pivotal study from the fall of 2023 has altered the perception of AI's deficiencies. The study specifically highlights the capabilities of generative AI — with ChatGPT 4.0 as the prime example — and demonstrates that AI can ideate with a sophistication akin to that of humans.



REMEMBER Please be aware that this level of ideation typically requires that human beings engineer precise prompting for the AI and provide domain-specific context to the large language model (LLM) involved (unless the model already possesses the relevant information). Nonetheless, the results of specific ideation studies have shown AI-generated ideas to be viable and on par with (if not better than) human-generated ones.



TECHNICAL STUFF **CHATGPT VERSUS WHARTON MBA STUDENTS**

Professors Christian Tardiest and Karl Ulrich gave 200 randomly selected MBA students from the University of Pennsylvania Wharton School of Business the task of generating an idea for a new product or service appealing to college students that a company could make available to those students for $50 or less. The professors gave the same assignment to ChatGPT 4, asking it to develop 100 ideas. To further the hypothesis, the professors asked ChatGPT 4 to develop an additional 100 ideas after some coaching, which involved providing ChatGPT 4 with a sample of successful ideas from past courses. After that exercise, they now had 400 ideas — 200 from MBA students and 200 from ChatGPT 4.

To assess the strength of the idea generation, the professors looked at the efficiency with which the ideas were generated per unit of time, the average quality of the ideas, and the number of truly exceptional ideas. Not surprisingly, ChatGPT 4 produced ideas very quickly — spitting out 200 ideas in about an hour (with some human supervision). The human-generated ideas took longer to develop. The study employed an online purchase-intent survey directed toward the targeted customer base to assess the quality of the ideas by asking the following question:

- "How likely would you be to purchase based on this concept if it were available to you?"

The possible responses ranged from *Definitely Wouldn't Purchase* to *Definitely Would Purchase.*

The professors got some very interesting results in the following average purchase probabilities:

- **Human-generated idea:** 40 percent
- **Uncoached ChatGPT 4 idea:** 47 percent
- **Coached ChatGPT 4 idea:** 49 percent

These results show that not only was ChatGPT 4 much faster in generating ideas but also that the overall quality of the ideas was stronger.

But, the study offered more results than just the purchase probability information. To determine whether students or AI produced exceptional ideas (after all, a business typically needs only one exceptional idea, rather than dozens of above-average ones), the study reviewed the top 10 percent of the ideas in each pool. Of these 40 ideas, a whopping 35 were created by ChatGPT 4, with 20 from the pre-trained ChatGPT 4 set and 15 from the untrained version of ChatGPT 4. ChatGPT 4 clearly came out on top in these results, as well, demonstrating that generative AI can not only perform basic copywriting tasks but also provide higher-order critical thinking and ideation.

The Wharton study, which included both MBA students and an AI tool, generated 400 product ideas (200 from students and 200 from ChatGPT 4). After market testing for the quality of all ideas, the average probability of purchase for human-generated ideas was 40 percent, whereas the average probability of purchase for the AI-generated ideas was 47 to 49 percent. The study also accounted for the speed of idea generation to note efficiency. (Unsurprisingly, ChatGPT 4 generated its ideas much faster than the students generated theirs.) Further review of the top 10 percent of ideas (40 ideas) revealed that AI generated 35 of those ideas.

This groundbreaking study highlighted the quality, quantity, and efficiency of AI-generated ideas and provided these important takeaways:

- **Generative AI can be an effective idea-generation tool.** Arguably, no piece of technology in the history of humankind has been able to achieve the kind of results demonstrated in the Wharton study. Using AI to generate ideas marks a distinct departure from the past uses of technology, which typically centered around knowledge management, automation, and calculations.
- **The role of human beings may shift in a world in which AI generates ideas.** Rather than only human beings serving as idea generators, our roles may now shift to becoming the prompters and evaluators of good ideas. The mechanism used for selecting ideas becomes even more important, and a mix of AI and humans (who have good taste) can best determine what ideas are most applicable in a specific business context.
- **The more human beings guide the AI, the better the AI gets as an idea generator.** Think of the AI like a copilot who can generate ideas with strong human inputs, rather than like an idea generator that competes with human beings to come up with the best ideas.
- **You need human-machine collaboration to deliver better products and services in the future.** The authors of the Wharton study pointed out this fact in their study described in this section. Human-AI collaboration at every stage of the idea-generation process can allow for stronger ideas than humans developing them alone.

If you still feel skeptical about the use of generative AI in ideation and consider the described Wharton study a theoretical exercise, take a look at these real-world examples:

- **Mattel harnessed AI to drive product development ideas for its Hot Wheels brand.** The company used generative AI to create four times as many product concept images as it normally would in the same amount of time. As a study from McKinsey & Company (a management consulting company) titled "How generative AI can boost consumer marketing" pointed out, this large number of concept images inspired new features and designs for products. You can find the study at `www.mckinsey.com/capabilities/growth-marketing-and-sales/our-insights/how-generative-ai-can-boost-consumer-marketing`.
- **Coca-Cola released a limited-edition soda flavor called Y3000, which was co-created with AI.** Coca-Cola used generative AI to analyze consumer flavor preferences so that it could design what turned out to be a raspberry slushy–like beverage that contained different artificial sweeteners. The company also used

AI to create futuristic artwork for the packaging. (The specific tool used to create the artwork was Stable Diffusion, from Stability AI [`https://stability.ai/`].)

- **Airbus applied AI-powered generative design and optimization to the A320neo aircraft.** By using generative design technology, Airbus optimized the A320neo's wingtip design, cutting the aircraft's fuel consumption by 3.5 percent, thereby setting a prime example of AI application in product design.
- **Adidas harnessed AI to design its Futurecraft 4D shoe.** The AI-influenced product design enhanced the shoe's sole comfort by including an internal lattice-like structure born out of the AI guidance. This design offers support and cushioning while maintaining durability and lightness.

Deciding Whether AI Hallucinations Are a Feature or a Bug

Inherent in large language models (LLMs; advanced AI systems trained on vast amounts of text data to generate human-like text) are *hallucinations,* the scenarios in which the LLMs generate incorrect, fabricated, or misleading information. This unique and relatively uncommon phenomenon causes the model to confidently present its results as accurate, even though the results may have no basis in reality.

Hallucinations may develop for a number of reasons, including

- **The vast data sets used to train LLMs** have gaps or inaccuracies, leading to incorrect generalizations.
- **The LLMs lack personal experience** and may be exposed to data that has limited timeframes. In these cases, LLMs don't get the benefit of learning from what happened outside their data set or processing parameters.
- **Inherent model limitations** related to statistical correlations or complex and ambiguous human-created prompts that can misdirect results.

On the surface, you can view hallucinations as a bug — a negative result to watch out for. I encourage you (and all users of generative-AI solutions) to apply common sense tests and verify outputs so that you can determine whether the AI is being factual. But when it comes to ideation, hallucinations have another side. You may view the capacity of the LLM to fabricate information as creative output that opens other possibilities for more novel outcomes.



REMEMBER By integrating generative AI into the creative process in marketing, you can unlock new types of ideas while also helping your human teammates confront their own biases. And when you layer in the opportunities driven by AI hallucinations, the potential benefits increase exponentially. So, hallucinations can actually help you with creative marketing.

Bringing in unexpected ideas and concepts

AI hallucinations can lead to unexpected, novel, and completely new ideas that break conventional thinking patterns (you know, break out of that box). These ideas likely happen because the AI engine isn't an entity grounded in real-world experience but, instead, is a text-driven statistical model.

Suppose that your marketing team uses an AI tool such as ChatGPT to brainstorm ideas for a new advertising campaign for a sports drink. Your team members input various prompts related to sports, energy, and refreshment. The AI, experiencing a hallucination, generates a concept about a futuristic sport that doesn't exist.

Instead of discarding this output as irrelevant, the team finds inspiration in the AI's mistake. The team members work together to develop a campaign around this imaginary sport, creating a series of drink ads that depict athletes training for this futuristic game. This kind of idea can capture the audience's imagination, setting the brand apart, and highlighting the drink's association with innovation, energy, and forward-thinking.

Branching out with non-traditional storytelling

You can harness AI-induced mistakes to create a new dimension for the brand's narrative by creating new stories that resonate with audiences in unexpected ways. This storytelling is particularly valuable in social media marketing because it captures attention, sparks engagement, and fosters a sense of authenticity and spontaneity.

Imagine that a travel agency uses AI to generate marketing content for promoting vacation packages. The AI, experiencing a hallucination, generates a description of a fictional, exotic destination that has unique cultural attributes and natural wonders that don't actually exist. Instead of dismissing this generation as an error, the marketing team sees an opportunity for an imaginative campaign. They develop a thematic campaign titled *Imagined Destinations,* which invites people to explore these AI-created places through a series of interactive online experiences, including virtual tours, digital art, and creative storytelling that showcase these fantastical destinations.

To increase engagement, the campaign includes an interactive element through which customers can submit their own ideas for imagined destinations, which the AI then expands upon. This user-generated content not only fosters community engagement but also provides insights into what types of vacations and destinations appeal to their clientele.

The campaign garners widespread attention for its creativity and use of technology, enhancing the travel agency's reputation for innovation. It also serves as a fun, engaging way for potential customers to dream about travel, keeping the agency top-of-mind when those could-be clients get ready to book real trips.

Facilitating testing and experimentation

You can put AI hallucination ideas into market research and test them in a similar fashion to the new ideas developed by the Wharton School study mentioned in the section "Engaging AI to Ideate on Behalf of Human Beings," earlier in this chapter. Even if these ideas don't lead to successful products, the exercise can show you what not to do or may encourage new ways of thinking for the marketing teams involved.

Suppose that a food company uses an AI model to explore new flavor ideas for a line of snacks. During a brainstorming session, the AI, because of a hallucination, suggests a bizarre and seemingly unpalatable flavor combination — something that no human being would consider.

Intrigued by the unusual suggestion, the product development team decides to experiment with this idea. They refine the concept and create a small batch of a snack that has this unique flavor. The snack turns out to be surprisingly appealing in taste tests. The flavor, positioned as a limited edition, generates buzz because of its novelty and the story of its AI-inspired origin. The snack gains attention, particularly among adventurous consumers and food enthusiasts, driving sales and enhancing the brand's image as innovative and daring.

Staying the course with generative AI

Over time, I believe that LLMs can deliver more accurate results with fewer hallucinations. Sam Altman, the CEO of OpenAI, said as much when discussing the future of ChatGPT. He fully expects generative AI to become more predictable, accurate, and — in his own words — "boring" over time. Altman's projections may be right, but the very nature of how generative AI works can allow for forced hallucinations or pushing the boundaries of conventional thinking via specifically chosen input data and carefully crafted prompts.



TIP You may generally view AI hallucinations as errors that you need to correct, but you can leverage them as a source of creative inspiration that can lead to unique and innovative marketing outcomes. When you explore specific ways to drive ideation and planning by using generative AI, keep in mind that the prompts you write to direct your AI models determine how imaginative the AI can be on your behalf.

Following Practical Steps for Idea Generation with AI

A common and useful scenario for using generative AI involves idea generation during a creative brainstorming process. You may instigate the brainstorming session to identify a new product, a customer value proposition, a marketing campaign, or even a broader go-to-market strategy. The extent of ideas that you can generate is limited only by the prompts

you articulate and the existing data or examples that you feed into the AI engine. In the following sections, I explain how you can use AI as a copilot when you run a combined human and AI ideation exercise.

Starting with the right prompts

A good starting point for combining human creativity with AI capabilities is to establish a prompt for both parties. Take, for example, a scenario in which you want to generate ten advertising concepts for an Apple ad promoting the latest iPhone during the Super Bowl. You can pose this challenge to both the human participants in the room and a generative AI engine such as ChatGPT. Here's what can happen:

- **The human team:** Draws on their personal experiences and knowledge about Apple and its iPhones to brainstorm ideas during a timeframe that you specify. Human understanding — informed by experiences, a sense of taste, and familiarity with what resonates with consumers (especially in the context of Super Bowl ads) — goes into the ideas that the team creates.
- **ChatGPT:** Generates ten diverse ideas in just a few seconds; some of the ideas, of course, are more compelling than others. ***Note:*** The richer the information that you provide to the AI, the more robust the ideas it generates. If you include Apple's brand strategy, recent research on the most desired iPhone features, and scripts from Apple's past successful ads, you can significantly enhance the AI's ideas.



REMEMBER You gather AI results from ChatGPT not to replace human creativity; those results may not match human capabilities. Instead, the real value emerges when the humans involved in the brainstorming session review the AI-generated ideas and select their top three or four favorites from both lists. This shortlist then forms the basis for crafting new, refined prompts, which you can use in subsequent iterations with ChatGPT to further develop and refine the chosen concepts.

This straightforward example illustrates a principle that you can apply to various types of idea-generating exercises. The approach works well, whether you want to focus your efforts on advertising, product feature definition, or strategic planning.

Stepping through an AI-for-ideation exercise

With information from the preceding section in mind, follow these steps when you use generative AI in an idea-generating exercise:

1. **Define the objectives of your ideation session.**
   Clearly define what you want to achieve by using AI-generated ideas. You may be after product innovation, marketing strategies, problem-solving for existing initiatives, or new creative marketing campaigns. As you set the objectives, also establish the boundaries and criteria for the ideas that you want, including
   1. *Feasibility:* How much time and effort can you spend on a resulting project?
   2. *Target audience:* Are you going after existing customers or branching out to a new target audience?
   3. *Budget constraints:* Do you have unlimited funds, or have they been established by previous projects?

   

   REMEMBER Setting objectives and boundaries is always a good business practice, but it's all the more important in the context of AI because you may have to face an even more divergent set of ideas than you do without AI input.
2. **Select the right AI tool(s) for your stated objectives.**
   Choose an appropriate AI model based on what you want to accomplish with your ideas. Depending on your needs, choose an AI model that specializes in idea generation in the format that fits your goals, such as
   1. ChatGPT (`http://chat.openai.com`) for text-based ideas
   2. DALL-E (`http://openai.com/dall-e`) for visual concepts
   3. Stable Diffusion (`http://stability.ai/stablediffusion`) and Runway (`http://runwayml.com`) for multi-modal idea generation

   In some cases, you may want to use a combination of tools — starting with ChatGPT and then taking the results into an image- or video-based tool after that. Doing this allows you to use each tool for what it is best at.
3. **Prepare and input data for your ideation session.**
   Compile data that's relevant to your objective. Examples include market trends, consumer behavior data, previous campaign results, competitor dynamics, and previous product launches. Provide the AI with as much context as possible to guide its idea generation process. As I note in the preceding section, the more quality data that you input, the better the potential outcomes.

   

   WARNING When selecting data for your idea-generating exercise, be careful about sharing sensitive information because it can lead to privacy breaches,

data misuse, and potential legal ramifications.

4. **Engage your AI tool to generate ideas.**
   Run your initial queries or prompts through the AI tool to start generating ideas. To stress test how strong your prompts are, have your human creative team respond to the same prompts. Doing so can quickly tell you whether you need to make the queries more specific because it will reveal gaps in clarity and precision. Adjusting the queries for the gaps you reveal can help ensure that the AI outputs are relevant and useful.



TIP When you run your prompts through your AI tool, specifically encourage the AI to generate a plethora of ideas based on different assumptions. Consider the advice about prompt engineering that you can find in Chapters 2 and 6 of Book 2 and encourage the AI to respond in the role of a creative director, a product designer, or a futurist, for example.

5. **Refine prompts based on resulting ideas and then try again.**
   Analyze the ideas generated by the AI for relevance, innovation, creativity, and feasibility. Use feedback loops in which you refine your queries based on initial outputs to guide the AI toward solutions aligned with your intent. And, as you do this, consider highlighting the best ideas and asking the AI to generate more like those ideas. The more guidance you give the AI, the better the subsequent outputs are.
6. **Evaluate and select viable ideas.**
   Use your predefined criteria (from Step 1) to evaluate and shortlist the most promising ideas. As you create this list, consider incorporating feedback not just from your creative team, but also from stakeholders in other departments or maybe even consumer groups. As the Wharton professors show in their research (see the sidebar "ChatGPT versus Wharton MBA students," in this chapter), online purchase-intent surveys can provide you with a powerful customer-centric tool for evaluating ideas.
7. **Develop and test your generated ideas.**
   For product ideas, you can develop prototypes. For concepts such as marketing strategies, you can create testable models or campaigns or take the ideas into your formal campaign briefing or creative development process. In many cases, you can use other AI tools to translate text-based ideas into conceptual drawings, prototypes, or video animations that you can then test through focus groups, pilot runs, or online surveys.

Deciding on AI Ideation Tools to Use

Recently, many AI ideation tools have cropped up. Some of these tools are standalone solutions, such as Adobe Firefly (`https://firefly.adobe.com/`) and Google's AutoDraw (`www.autodraw.com`); other tools are integrated into broader software suites.

For example, TikTok's Creative Assistant — which launched in September 2023 — is particularly notable. Situated in TikTok's Creative Center (`http://ads.tiktok.com/business/creativecenter`), this AI-driven virtual assistant offers advertisers a dynamic toolbox for content creation. By tapping into TikTok-specific creative insights, the Creative Assistant offers relevant suggestions and guidance that facilitate the crafting of compelling and effective content for TikTok campaigns.

Creative Assistant acts as a virtual collaborator, helping you in various facets of content creation, such as

- Guiding newcomers through best practices on the TikTok platform
- Displaying and analyzing top-performing ads to aid in creative research
- Brainstorming ideas and refining scripts collaboratively, especially when you're struggling to develop new ideas



REMEMBER The Creative Assistant tool augments human creativity, symbolizing the fusion of human ingenuity with AI capabilities. It empowers creators to continually enhance their presence on TikTok through perpetual ideation sessions. But although the Creative Assistant offers suggestions and collaborates in the creation journey, the creator retains absolute control over the materials, having the liberty to use, edit, discard, or redo them, as desired.

You have literally hundreds of AI tools to consider — many of which use ChatGPT as a foundation — and Table 1-1 highlights some notable ones. Although some of the tools in this table are more oriented toward content creation rather than idea generation, they can still play the role of idea generators, too.

TABLE 1-1 AI Tools for Idea Generation

Tool	*Website*	*What It Is*	*What It Does*
ContextMinds	`www.context-minds.com`	A tool that assists in research and idea generation, combining mind-mapping functionality with AI-driven content suggestions	Organizes thoughts and makes connections through relevant articles, papers, and videos related to their topics of interest
HyperWrite	`www.hyper-writeai.com`	A writing assistant that helps users develop ideas and refine original text	Generates various writing formats such as e-mails or marketing copy
Jasper	`www.jasper.ai`	A tool that assists in generating written content for	Crafts compelling narratives and ideas for marketing copy, blog posts, and so on

Tool	*Website*	*What It Is*	*What It Does*
		various purposes	
Pictory	http://pictory.ai	A tool for ideation in video marketing, transforming existing content into engaging, shareable video formats	Creates short videos from long-form content such as webinars or blog posts
Riffusion	www.riffusion.com	A tool that synthesizes music based on AI-generated sound patterns	Experiments with unique soundscapes and auditory branding elements
Runway	http://runwayml.com	A platform for experimenting with machine learning (ML) mod-	Offers ideation in the realms of design, art, and multimedia, to explore new visual con-

Tool	*Website*	*What It Is*	*What It Does*
		els in real time	cepts and effects
Stable Diffusion	http://stability.ai	A text-to-image model that allows users to create visual images from textual descriptions	Helps artists, designers, and marketers to rapidly generate diverse and creative visual concepts
Synthesia	www.synthesia.io	A video creation platform that generates high-quality synthetic media and personalized video content	Creates realistic digital avatars and voiceovers, streamlining the video production process
Wondershare EdrawMind	http://edrawmind.wondershare.com	A tool that facilitates organized visual represen-	Helps create mind maps, flowcharts, and organizational charts

Tool	*Website*	*What It Is*	*What It Does*
		tation of ideas while also helping to generate ideas	to enhance collaborative thinking and planning
Writer	http://writer.-com	A writing assistant for busi-nesses that helps create clear, con-sistent, and brand-aligned content	Assists with grammar, style, and tone to ensure high-quality writing across all company communica-tions